

UNIT-IV B

Classical free Electron Theory (Drude Theory)

→ when we apply electric field between two ends of metal, then the force acting on e^- is given by

$$f = eE \quad \text{--- (1)}$$

according to Newton's law

$$F = ma \quad \text{--- (2)}$$

$$ma = eE$$

$$a = \frac{eE}{m} \quad \text{--- (3)}$$

In the presence of electric field drift velocity of e^- is

$$V_d = a\tau \quad \text{--- (4)}$$

V_d = drift velocity

a = acceleration

using eq (3)

τ = collision time

$$V_d = \frac{eE\tau}{m} \quad \text{--- (5)}$$

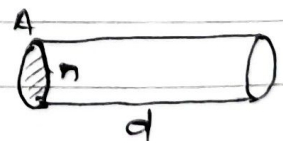
Hence, collision time τ is given by

$$\tau = \frac{\lambda}{\bar{v}}$$

λ = collision length

$\bar{v}$ = mean square speed = $\sqrt{\frac{3kT}{m}}$

$$V_d = \frac{eE}{m} \cdot \frac{\lambda}{\bar{v}} \quad \text{--- (6)}$$



→ for a metal having length 'd' and area of cross section 'A' total number of charge on the metal is $= neAd$

Altho current I is

$$I = \frac{\text{charge}}{\text{time}} = \frac{neAd}{t}$$

$$I = neAv_d \text{ --- (7)} \quad \therefore v_d = \frac{d}{t} = \text{drift velocity}$$

Now the current density J is between two ends of given by

$$J = \sigma E, \quad \sigma = \text{conductivity}$$

$$\frac{I}{A} = \sigma E$$

using eq (7)

$$\frac{neAv_d}{A} = \sigma E$$

$$nev_d = \sigma E$$

using eq -- (6)

$$ne \frac{e^2 \lambda}{m v} = \sigma E$$

$$\boxed{\frac{ne^2 \lambda}{m v} = \sigma} \text{ --- (8)}$$

$$\text{Put } v = \sqrt{\frac{3KT}{m}}$$

$$\frac{ne^2 \lambda}{m} \frac{\sqrt{m}}{\sqrt{3KT}} = \sigma$$

$$\frac{ne^2 \lambda \sqrt{m}}{\sqrt{m^2 3KT}} = \sigma$$

$$\sigma = \frac{ne^2 \lambda}{\sqrt{3} m k T} \quad \text{--- (8)}$$

conductivity.

$$\text{Resistivity } \rho = \frac{1}{\sigma}$$

$$\rho = \frac{m \bar{v}}{ne^2 \lambda} \quad \text{--- (9)}$$

$$\rho = \frac{\sqrt{3} m k T}{ne^2 \lambda} \quad \text{--- (9)}$$

$$\text{mobility } \mu = \frac{V_d}{E} \quad \text{--- (10)}$$

$$V_d = \frac{e E \tau}{m} \quad \text{--- using eq (5)}$$

Put in eq (10)

$$\mu = \frac{e E \tau}{m} \cdot \frac{1}{E}$$

$$\mu = \frac{e \tau}{m}$$

$$\tau = \frac{m \mu}{e} \quad \text{--- (11)}$$

Since,

$$\sigma = \frac{ne^2}{m} \frac{\lambda}{v} = \frac{ne^2 \tau}{m}$$

using eq (11)

$$\sigma = \frac{ne^2}{m} \cdot \frac{m \mu}{e}$$

$$\sigma = ne \mu$$

This is the relation between conductivity and mobility.

when a potential difference V is apply

$$R = \frac{\rho l}{A} \quad \Rightarrow \quad \rho = \frac{RA}{l}$$

$$E = \frac{V}{l}$$

Now,

$$J = \sigma E$$

$$\frac{I}{A} = \frac{E}{\rho} \quad \text{--- is equivalent}$$

$$\frac{I}{A} = \frac{V}{l} \cdot \frac{l}{RA}$$

$$\boxed{V = IR} \rightarrow \text{Ohm's law}$$

Quantum free electron theory (same as particle in 1-Dimensional Box)

→ Electrons are moving freely inside the box of length 'L'.

$$\Psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \quad n = 1, 2, 3, \dots$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad \hbar = \frac{h}{2\pi}$$

fermi Energy

→ The energy ~~upto~~ below which energy states are filled with electrons and above which energy states are empty at 0K is called fermi energy or fermi level.

Ques: Derive an expression for density of state and fermi-dirac distribution. (Previous year)

Ans: D(E) Density of states defined as number of electronic states per unit energy range.

$$\text{Density of states } D(E) = \frac{2dn}{dE} \quad \text{--- (1)}$$

Here, 2 is written for two spins of e^-

Energy of e^- in n^{th} state is

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad \text{--- (2)}$$

Take derivative both sides

$$de = \frac{2n\pi^2\hbar^2}{2ml^2} dn$$

$$de = \frac{n\pi^2\hbar^2}{ml^2} dn$$

$$\boxed{\frac{dn}{de} = \frac{ml^2}{n\pi^2\hbar^2}}$$

Put this value in eq (2)

$$D(e) = \frac{2ml^2}{n\pi^2\hbar^2} \quad (3)$$

Using (2)

$$n^2 = \frac{2ml^2 E_n}{\pi^2 \hbar^2}$$

$$n = \left(\frac{2ml^2}{\pi^2 \hbar^2} \right)^{1/2} E^{1/2}$$

Put in (3)

$$D(E) = \frac{2ml^2}{\pi^2 \hbar^2} \left(\frac{\pi^2 \hbar^2}{2ml^2} \right)^{1/2} \cdot \frac{1}{E^{1/2}}$$

$$= \frac{2ml^2}{\pi^2 \hbar^2} \cdot \frac{\pi \hbar}{l} \left(\frac{1}{2m} \right)^{1/2} \frac{1}{E^{1/2}}$$

$$= \frac{(2m)L}{\pi \hbar} (2m)^{-1/2} \cdot \frac{1}{E^{1/2}}$$

$$D(E) = \frac{L}{\pi \hbar} \frac{(2m)^{1/2}}{E^{1/2}}$$

$$D(E) \propto E^{-1/2}$$

U.V.I

Fermi-Dirac Distribution

→ It is given by —

$$F(E) = \frac{1}{1 + e^{(E - E_f)/k_B T}}$$

E_f = Fermi Energy

k_B = Boltzmann constant

T = Temperature

(i) at 0 K, $E < E_f \Rightarrow E - E_f = -ve$

$$F(E) = \frac{1}{1 + e^{-ve/0}} = \frac{1}{1 + e^{-\infty}} = \frac{1}{1 + e^{\frac{1}{\infty}}}$$

$$e^{\infty} = \infty$$

$$F(E) = \frac{1}{1 + \frac{1}{\infty}} = \frac{1}{1 + 0} = 1$$

(ii) at 0 K, $E > E_f$

$$F(E) = \frac{1}{1 + e^{+ve/0}}$$

$$= \frac{1}{1+e^\infty} = \frac{1}{1+\infty} = \frac{1}{\infty}$$

$$F(E) = 0$$

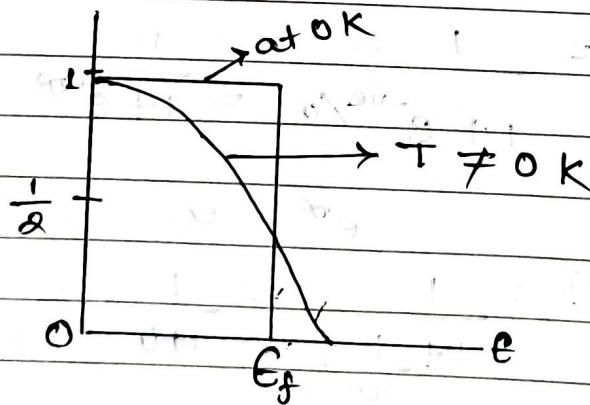
(iii) at 0 K, $E = E_f \Rightarrow E - E_f = 0$

$$F(E) = \frac{1}{1+e^0} = \frac{1}{1+1} = \frac{1}{2}$$

(iv) $E > E_f$, $T \neq 0$ K, $E - E_f = +ve$

$$F(E) = \frac{1}{1+e^{+ve/T}}$$

$$\neq 0$$



failure or limitation of Quantum free

- It is unable to distinguish between insulator, semiconductor and conductor.
- It is not able to explain why some solids are conductors.
- It doesn't explain metallic properties of crystals.
- It is unable to explain positive value hall co-efficient.

Thermionic Emission and Richardson's Equation

- let us suppose that motion of electron is along X-axis.
- The phenomena of emission of electrons from the surface of metal, when the metal is heated is called thermionic emission.
- It will occur when the K.E of electron along X-axis.

$$\frac{1}{2} m v^2 \geq E_f + \phi \quad \phi = \text{work function}$$

$$\text{Minimum energy } \frac{1}{2} m v_x^2 = E_f + \phi$$

$$\boxed{v_x^2 = \frac{2(E_f + \phi)}{m}} \quad \text{minimum} \quad \dots (1)$$

The density of states of electrons per unit volume in the momentum range " p to $p+dp$ " is given by

$$N = \frac{8\pi p^2 dp}{h^3 [1 + e^{(E - E_f)/k_B T}]} \quad \text{--- (2)}$$

p = linear momentum

$$p = mv$$

$$dp = m dv$$

Put these value in (2)

$$N = \frac{8\pi m^2 v^2 m dv}{h^3 [1 + e^{(E - E_f)/k_B T}]}$$

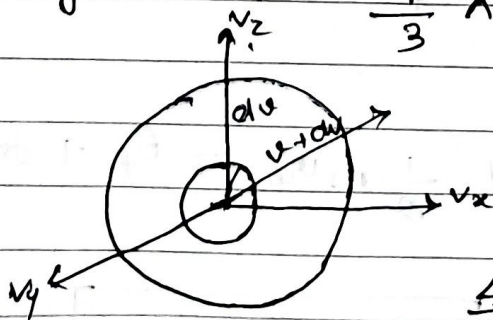
$$N = \frac{8\pi m^3 v^2 dv}{h^3 [1 + e^{(E - E_f)/k_B T}]} \quad \text{--- (3)}$$

If v_x , v_y and v_z are x , y and z components of velocity then the space velocity of electrons in 3D in the range v_x to $v_x + dv_x$

$$v_y \text{ to } v_y + dv_y$$

$$v_z \text{ to } v_z + dv_z$$

is given by $= \frac{4}{3} \pi [(v + dv)^3 - v^3]$



$$\frac{4}{3} \pi v^3$$

$$\frac{4}{3} \pi (v + dv)^3 - \frac{4}{3} \pi v^3$$

$$\begin{aligned}
 dv_x dv_y dv_z &= \frac{4}{3} \pi [(v+dv)^3 - v^3] \\
 &= \frac{4}{3} \pi [v^3 + dv^3 + 3dv(v+dv) - v^3] \\
 &= \frac{4}{3} \pi [(dv^3) + 3v^2 dv + 3v(dv^2)]
 \end{aligned}$$

If dv is very small $\Rightarrow dv^2$ and dv^3 are very very small.

So, we can neglect dv^2 and dv^3

$$\begin{aligned}
 dv_x dv_y dv_z &= \frac{4}{3} \pi \times 3v^2 dv \\
 &= 4\pi v^2 dv
 \end{aligned}$$

$$v^2 dv = \frac{dv_x dv_y dv_z}{4\pi}$$

put in (3)

$$N = \frac{2 \pi m^3}{h^3 \times 4\pi} \frac{dv_x dv_y dv_z}{[1 + e^{(E-E_f)/k_B T}]}$$

$$N = \frac{2m^3}{h^3} \frac{dv_x dv_y dv_z}{[1 + e^{(E-E_f)/k_B T}]} \quad \text{--- (4)}$$

When we heat the metal surface, then E will become very large i.e., $E - E_f$ and hence $(E - E_f)/k_B T$ will become very large.

Hence, we can neglect 1 in comparison to $e^{(E-E_f)/k_B T}$

$$N = \frac{2m^3}{h^3} \frac{dv_x dv_y dv_z}{e^{(E-E_f)/k_B T}} = \frac{2m^3}{h^3} \frac{dv_x dv_y dv_z}{e^{(E/k_B T - E_f/k_B T)}}$$

$$N = \frac{2m^3}{h^3} e^{-E/k_B T} e^{E_f/k_B T} dx dy dz$$

$$\text{Put } A = \frac{2m^3}{h^3} e^{E_f/k_B T}$$

$$N = A(e)^{-E/k_B T} dx dy dz$$

The current density of electrons emitted along x-axis is given by $J_x = Ne v_x$

$$J_x = A e^{-E/k_B T} e v_x$$

$$E_x = \frac{1}{2} m v_x^2$$

$$E_y = \frac{1}{2} m v_y^2$$

$$E_z = \frac{1}{2} m v_z^2$$

$$J_x = A e^{\frac{-1}{2} \frac{m v_x^2}{k_B T} - \frac{1}{2} \frac{m v_y^2}{k_B T} - \frac{1}{2} \frac{m v_z^2}{k_B T}}$$

$$J_x = A \int v_x e^{\frac{-1}{2} \frac{m v_x^2}{k_B T}} dx \int e^{\frac{-1}{2} \frac{m v_y^2}{k_B T}} dy \int e^{\frac{-1}{2} \frac{m v_z^2}{k_B T}}$$

Imarks

$$J_x = A T^2 e^{-\Phi/k_B T} \rightarrow \text{Richardson's Equation}$$

$\therefore J_x = \text{Current density.}$

UNIT-IV A

Date: _____

Crystal Structure

Lattice parameter of a unit cell.

- ① distance in all direction. $x, y, z \rightarrow$ axis
- ② angle α, β, γ

Bravais's Lattice

→ The lattice where in which every lattice point has identical surrounding and obeys symmetric operation is called Bravais's Lattice.

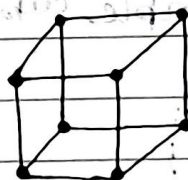
There are five Bravais's lattice in 2-D and there are fourteen Bravais's lattice in 3-D.

2-D

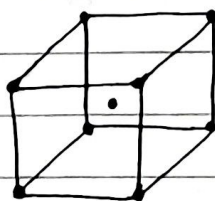
- $a = b, \gamma = 90^\circ$ square
- $a = b, \gamma = 120^\circ$ 60° rhombus
- $a \neq b, \gamma \neq 90^\circ$ parallelogram
- $a \neq b, \gamma = 90^\circ$ rectangle

3-D

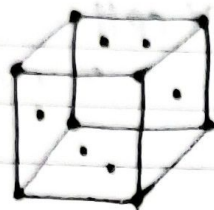
① simple cubic (SCC)



② Body Centred cubic (BCC)



⑧ face centred cube (fcc)



No of atoms per unit cell

$$N = \frac{N_c}{8} + \frac{N_b}{1} + \frac{N_f}{2}$$

$$① \text{ SSC } N = \frac{8}{8} + 0 + 0 = 1$$

$$② \text{ BCC } N = \frac{8}{8} + \frac{1}{1} + 0 = 2$$

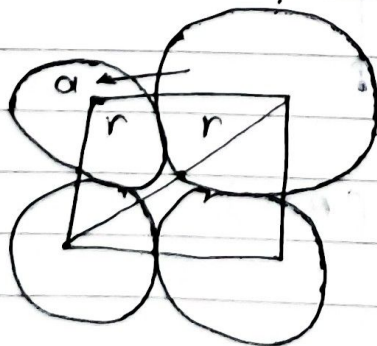
$$③ \text{ fcc } N = \frac{8}{8} + 0 + \frac{6}{2} = 1 + 3 = 4$$

Ques. Define the term bravais's lattice, crystal,

Atomic Radius :- It is defined as the distance between two nearest neighbouring atoms.

Atomic Radius

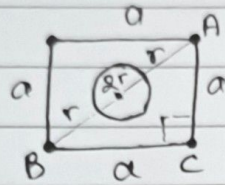
① Atomic Radius of simple cubic



$$2r = a$$

$$r_{sc} = \frac{a}{2}$$

(b) fcc



$$(AB)^2 = BC^2 + AC^2$$

$$(4r)^2 = a^2 + a^2$$

$$4r = \sqrt{2a^2}$$

$$r = \frac{\sqrt{2}a}{4}$$

$$r = \frac{\sqrt{2}}{4} \times \frac{\sqrt{2}}{\sqrt{2}} a = \frac{2a}{2\sqrt{2}} = \frac{a}{\sqrt{2}}$$

The no. of neighbouring atoms that surround every atom in the lattice is called co-ordination number.